9 In this question you must show detailed reasoning.

The curve $xy + y^2 = 8$ is shown in Fig. 9.

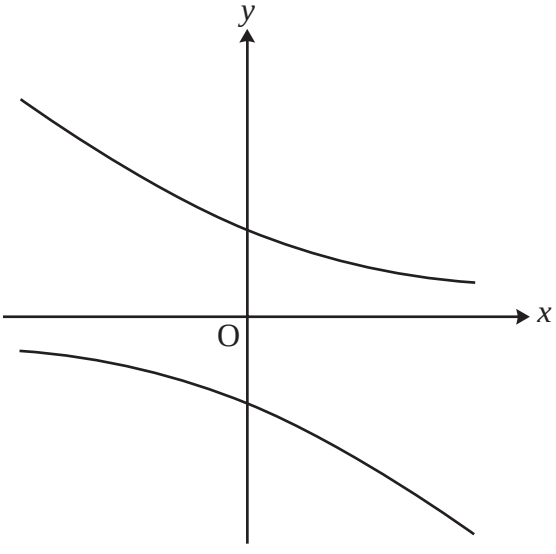

Fig. 9

Find the coordinates of the points on the curve at which the normal has gradient 2. **[6]**